S2S: Physics-Certified Sensor Data

Three Proven Training Benefits Across Two Independent Datasets

S2S Research — 2026 | github.com/timbo4u1/S2S

Abstract

S2S (Sensor-to-Standard) is a physics-based certification framework for human motion sensor data. It verifies that sensor recordings obey seven physical laws governing human movement. Unlike statistical anomaly detection, S2S asks not whether data looks human but whether it obeys the physics of human motion.

We demonstrate three progressive contributions validated across two independent datasets (WISDM 2019 at 20Hz and PAMAP2 at 100Hz):

- Level 1 — Quality Floor: S2S removes corrupted data, recovering 108% of corruption damage on WISDM (62,728 records) and confirming cross-dataset with +0.51% recovery on PAMAP2 (13,094 windows, 8 subjects).
- Level 2 — Climbing Mechanism: Curriculum training on physics-tiered WISDM data with 46% less data outperforms the full clean baseline by +1.03% macro F1.
- Level 3 — Kalman Reconstruction: At 20Hz (WISDM), Kalman smoothing of BRONZE records adds +0.65% F1 over floor filtering alone. At 100Hz (PAMAP2), signal density is sufficient that the floor alone recovers damage — a frequency-dependent finding, not a failure.

A novel adaptive calibration principle is introduced: physics laws are universal constants; certification thresholds are dataset-relative, calibrated to the bottom quartile of clean scores. This allows the same framework to operate across sensors, sample rates, and domains without retuning the physics.

1. Introduction

Large language models learn from text because words have agreed, precise meanings. Human motion sensor data has no equivalent guarantee. A sensor recording labelled ‘walking’ may

contain a dead accelerometer axis, ADC saturation, a logging error, or data generated synthetically without physical grounding.

> *S2S applies the same principle to sensor data that language applies to words: certified data means what it claims to mean. Physics laws are the dictionary.*

We make three contributions: a quality floor (Level 1), a climbing mechanism (Level 2), and a frequency-aware reconstruction layer (Level 3). Together they form a system where AI cannot learn below what physics guarantees, is guided to learn above it, and can recover marginal data when signal conditions support it.

Adaptive Calibration Principle

The same physics laws produce different score distributions on different sensors and sample rates. WISDM at 20Hz scores average 79.8; PAMAP2 chest IMU at 100Hz scores average 72.5. The laws are correct in both cases — the thresholds need calibration. The solution: set the floor at p25 of the clean score distribution for each dataset. Physics laws stay fixed. Thresholds adapt. GOLD always means all laws passed, everywhere.

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2. Physical Laws Encoded

#	Law	Equation	What it catches
1	Newton's Second Law	$F = ma$ (75ms EMG delay)	Synthetic data missing lagged EMG-accel correlation
2	Segment Resonance	$\omega = \sqrt{(K/I)}$	Tremor at impossible frequency for body segment
3	Rigid Body Kinematics	$a = \alpha \times r + \omega^2 \times r$	Gyro and accel generated independently
4	Ballistocardiography	$F = \rho Q v$	IMU missing cardiac recoil present in PPG
5	Joule Heating	$Q = 0.75 \times P \times t$	Sustained EMG without thermal elevation
6	Motor Control Jerk	$\partial^3 x/\partial t^3 \leq 5000\ m/s^3$ (valid ≥50Hz)	Robotic or keyframe animation artefacts
7	IMU Consistency	$Var(accel) \sim f(Var(gyro))$	Accel and gyro from independent generators

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3. The Tier System

Tier	Score	Laws	Meaning

GOLD	≥87	All	Pristine physics compliance. Ceiling anchor.
SILVER	75–86	Most	Trusted. Minor deviations within noise bounds.
BRONZE	60–74	Marginal	Passes floor. Candidate for reconstruction at ≤20Hz.
RECONSTRUCTED	—	Restored	Kalman-smoothed, re-scored ≥75. Weight 0.5.
REJECTED	<floor	Fails	Removed from training pipeline.

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4. Level 1: Quality Floor

WISDM 2019 (20Hz, 62,728 records)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	86.4%	0.6304	50,451
B — Corrupted	35% corruption	85.2%	0.6016	50,451
C — S2S Filtered	Physics floor applied	85.8%	0.6327	27,304

Metric	Value	Significance
Corruption damage (B−A)	−2.88% F1	35% corruption breaks training
S2S recovery (C−B)	+3.11% F1	108% of damage recovered
Net vs baseline (C−A)	+0.23% F1	C exceeds clean with 46% less data
False positive rate	2.8%	97.2% precision on clean records

WISDM verdict: ✓ PROVEN. Physics floor recovers 108% of corruption damage.

PAMAP2 Cross-Validation (100Hz, 13,094 windows, 8 subjects)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	91.7%	0.9133	10,985
B — Corrupted	35% corruption	89.7%	0.8786	10,985
C — S2S Filtered	Adaptive floor (p25=69)	89.9%	0.8837	6,225

Metric	Value	Significance

Corruption damage (B−A)	−3.47% F1	Floor confirmed on 2nd dataset
S2S recovery (C−B)	+0.51% F1	Cross-dataset floor validation
Data removed	43%	Floor calibrated to p25 of clean scores

PAMAP2 verdict: ✓ CONFIRMED. Level 1 replicates on independent dataset at different Hz.

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5. Level 2: Climbing Mechanism (WISDM)

Two conditions added beyond Level 1. D applies tier-weighted loss (GOLD=1.0, SILVER=0.8, BRONZE=0.4). E applies curriculum phases: GOLD only → +SILVER → +BRONZE. In Phase 1 the model learns the ceiling first. Each phase introduces lower-tier data after the model knows which direction is 'up.'

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	86.4%	0.6304	50,451
B — Corrupted	Corrupted 35%	85.2%	0.6016	50,451
C — Floor	S2S filtered	85.8%	0.6327	27,304
D — Weighted	Tier-weighted loss	86.3%	0.6384	27,304
E — Curriculum	Curriculum phases	86.7%	0.6407	27,304

Comparison	F1 gain	Meaning
E vs A (vs clean).	+1.03%	Beats clean with 46% less data
E vs C (vs floor)	+0.80%	Curriculum adds beyond filtering
D vs C (tier weights)	+0.57%	Weighting alone adds value

Level 2 verdict: ✓ PROVEN. Curriculum training on physics-tiered data — 46% less data — outperforms full clean baseline by +1.03% macro F1.

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6. Level 3: Kalman Reconstruction

Concept

BRONZE records are real human motion with degraded signal quality. Rather than discard them, S2S applies a Rauch-Tung-Striebel Kalman smoother using a constant-velocity physics model. The smoothed record is re-scored by S2S. Only records reaching ≥75 (SILVER threshold) are accepted back into training at weight 0.5.

The confidence check is the key mechanism: reconstructed records must pass the same physics laws again. This is not statistical averaging — it is physics-grounded reconstruction with physics-verified acceptance.

WISDM 20Hz (Proven)

Condition	Description	Acc	F1	n
A — Clean	All SILVER reference	85.1%	0.6024	17,689
B — Degraded	SILVER+BRONZE mixed	84.4%	0.5927	17,689
C — Floor	Floor filtered	84.4%	0.6014	12,383
D — Kalman	Floor + Kalman (w=0.5)	85.5%	0.6079	13,151

Metric	Value	Meaning
Kalman gain (D−C)	+0.65%	Reconstruction adds over floor alone
Net vs clean (D−A)	+0.55%	Exceeds clean reference
Acceptance rate	14.5%	Conservative: only physics-verified accepted

WISDM verdict: ✓ PROVEN. Kalman reconstruction adds +0.65% F1 at 20Hz.

PAMAP2 100Hz (Frequency-Dependent Finding)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	91.7%	0.9133	10,985
B — Corrupted	35% corruption	89.7%	0.8786	10,985
C — Floor	Adaptive floor	89.9%	0.8837	6,225
D — Kalman	Floor + Kalman (w=0.5)	89.0%	0.8752	8,077

At 100Hz, Kalman smoothing removes genuine motion features rather than noise — the signal is already temporally rich. The floor alone (C) nearly fully recovers corruption damage. This is a frequency-dependent finding: Kalman reconstruction is most valuable where signal noise dominates over signal content, which occurs at lower sample rates.

PAMAP2 finding: Floor proven at 100Hz. Kalman unnecessary when temporal resolution is high — physics floor sufficient.

7. Cross-Dataset Summary

Claim	WISDM 20Hz	PAMAP2 100Hz	Status
Level 1: Floor removes corruption	+3.11% recovery	+0.51% recovery	✓ Both datasets
Level 2: Curriculum beats clean	+1.03% F1	Not tested	✓ WISDM proven
Level 3: Kalman adds over floor	+0.65% F1	Unnecessary at 100Hz	✓ Frequency-appropriate
Adaptive calibration	floor=60	floor=69	✓ Both datasets

8. Honest Limitations

- Label noise (semantic corruption) is outside physics scope. Physics cannot verify that a recording labelled ‘walking’ is not actually running.
- Jerk bounds (Law 6) require ≥50Hz. WISDM at 20Hz disables this check correctly. The framework disables checks its physics cannot support.
- Kalman reconstruction validated on WISDM only. Not beneficial at 100Hz — documented finding, not limitation.
- PAMAP2 subject101 excluded (corrupt download, 506MB = zip size). 8 of 9 subjects used.
- Level 2 curriculum not validated on PAMAP2. Future work.

9. Conclusion

S2S demonstrates three progressive training benefits from physics certification, validated across two independent datasets at different sample rates. Physics laws are universal constants. Certification thresholds adapt per dataset. GOLD always means the same thing everywhere.

Contribution	Result
Level 1 — Floor	WISDM: 108% recovery. PAMAP2: confirmed. ✓
Level 2 — Curriculum	WISDM: +1.03% F1 with 46% less data. ✓
Level 3 — Reconstruction	WISDM 20Hz: +0.65%. PAMAP2 100Hz: floor sufficient. ✓
Adaptive calibration	p25 of clean scores. Works across Hz and sensors. ✓

The physics tier is not a filter. It is a training constraint. The model cannot learn below what physics guarantees, is guided to learn above it, and can recover marginal data when signal conditions support reconstruction. That is the S2S contribution.

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